# EE3621: Digital Signal Processing

Lecture 29: DSP for Radar — Pulse Compression, Doppler Processing & CFAR

(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

## 1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Radar Fundamentals: Range, Doppler, and Resolutions.
- **05:00 – 12:00 (7 mins):** Matched Filter & LFM (Chirp): Waveform equations and Pulse Compression.
- **12:00 – 18:00 (6 mins):** Pulse Compression Implementation: Stretch processing & FFT.
- **18:00 – 25:00 (7 mins):** Range-Slow Time Matrix & Doppler Processing: MTI & Pulse-Doppler.
- **25:00 – 30:00 (5 mins):** Constant False Alarm Rate (CFAR) Detection.
- **30:00 – 35:00 (5 mins):** FMCW Radar & SAR Basics.
- **35:00 – 40:00 (5 mins):** Checkpoints & Review.

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## 2 Radar Fundamentals

### 2.1 Range Equation

Radar transmits an electromagnetic pulse and waits for the echo. Let the time-of-flight of the pulse be $\tau$. The round-trip distance is $c\tau$, so the target range $R$ is:

$$R = \frac{c\tau}{2} \tag{1}$$

### 2.2 Doppler Frequency

If a target is moving with radial velocity $v$, the round trip distance changes as $2vt$, introducing a phase shift $\phi(t) = -\frac{2\pi(2vt)}{\lambda}$. The Doppler frequency shift is:

$$f_d = -\frac{1}{2\pi}\frac{d\phi}{dt} = \frac{2v}{\lambda} \tag{2}$$

### 2.3 Resolutions

The range resolution $\delta R$ and Doppler resolution $\delta v$ are:

$$\delta R = \frac{c}{2B} \tag{3}$$

$$\delta v = \frac{\lambda}{2T_{dwell}} \tag{4}$$

# 3 Matched Filter for Radar

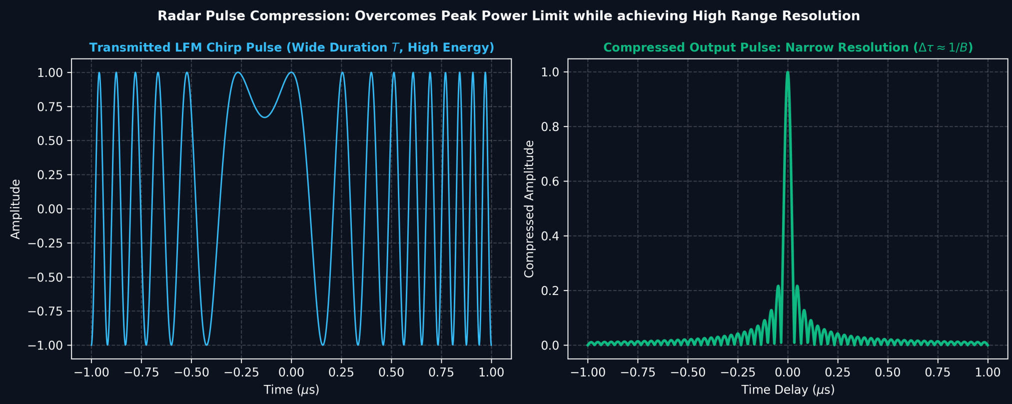


Figure 1: Radar Linear Frequency Modulation (Chirp) Pulse Compression for High Range Resolution.

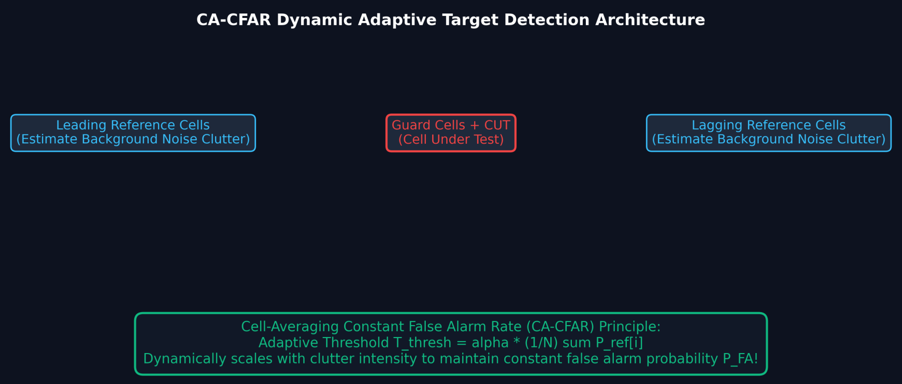


Figure 2: Cell-Averaging Constant False Alarm Rate (CA-CFAR) Adaptive Threshold Target Detection.

## 3.1 Linear Frequency Modulated (LFM) Chirp

The LFM chirp signal is defined as:

$$s(t) = \text{rect}(t/T)e^{j\pi\mu t^2} \tag{5}$$

where $\mu = B/T$ is the chirp rate.

## 3.2 Matched Filter Response

The matched filter impulse response is:

$$h(t) = s^*(-t) \tag{6}$$

The output pulse compression is:

$$|R_{ss}(\tau)|^2 = T^2\text{sinc}^2(\pi B\tau) \tag{7}$$

The Time-Bandwidth Product (TBP) represents the compression ratio:

$$\text{TBP} = B \cdot T \tag{8}$$

# 4 Pulse Compression Implementation

We use the Frequency Domain for fast processing:

$$Y(f) = R(f)H^*(f) = R(f)S^*(f) \tag{9}$$

This is implemented as an FFT-Multiply-IFFT structure. For extreme bandwidths, stretch processing mixes the signal with a reference chirp.

# 5 Range Cell Processing

A radar transmits $N_{chirps}$ pulses. We stack each received pulse into a 2D **Range-Slow Time Matrix**, where rows are range bins and columns are successive pulses.

# 6 Doppler Processing (MTI & Pulse-Doppler)

## 6.1 Moving Target Indicator (MTI)

MTI subtracts consecutive pulses to cancel stationary clutter:

$$y(n) = x(n) - x(n-1) \tag{10}$$

In designing more complex MTI filters, analog prototype filters can be digitized using the Matched Z-Transform (MZT) (Figure 1).

Spectral transformations can adapt the MTI notch (Figure 2).

## 6.2 Pulse-Doppler Processing

A 2D FFT over the Range-Slow Time Matrix produces a **Range-Doppler Map**, providing both distance and velocity.

# 7 Constant False Alarm Rate (CFAR) Detection

The threshold is $T_{thresh} = T \cdot \bar{P}_{noise}$. For Cell Averaging CFAR (CA-CFAR) with $N$ reference cells and false alarm probability $P_{fa}$, the multiplier is:

$$T = N\left(P_{fa}^{-1/N} - 1\right) \tag{11}$$

# 8 FMCW Radar & SAR Basics

## 8.1 FMCW Radar

The beat frequency for a stationary target is:

$$f_b = \frac{2R}{c}\frac{B}{T} \tag{12}$$

## 8.2 SAR Basics

Synthetic Aperture Radar produces high-resolution images. The azimuth resolution is surprisingly independent of range:

$$\delta a = \frac{D}{2} \tag{13}$$

## 9 Checkpoint & Quick Review Questions

**Q1:** Calculate range resolution and TBP for $B = 50$ MHz and $T = 20\mu s$.
*Answer:* $\delta R = \frac{c}{2B} = 3$ m. $TBP = 50 \times 10^6 \times 20 \times 10^{-6} = 1000$.

**Q2:** Find CFAR threshold $T$ for $N = 16$ and $P_{fa} = 10^{-6}$.
*Answer:* $T = 16((10^{-6})^{-1/16} - 1) = 16(10^{0.375} - 1) \approx 21.94$.

**Q3:** An FMCW radar with $B = 1$ GHz and $T = 100\mu s$ has a beat frequency of 150 kHz. Find range $R$.
*Answer:* $R = \frac{f_b c T}{2B} = \frac{150\times10^3\times3\times10^8\times100\times10^{-6}}{2\times10^9} = 2.25$ m.